



Task Scheduler

Work Tracker & Project Manager

Complete User Guide — All Features

 task-scheduler-production-39cb.up.railway.app

Built with Spring Boot 3 · Tailwind CSS · Spring Security
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1 Registration & Login

Creating an Account

1. Go to `/register` or click **"Create one"** on the login page
 2. Fill in all four fields: Username, Email, Password, Confirm Password
 3. Click **Register**
 4. On success → redirected to login with a confirmation message
 5. Login with your new credentials

Registration Validation Rules

Field	Rule	Error Message
Username	Required, 3–30 characters	"Username must be 3-30 characters"
Username	Must be unique (not already taken)	"Username already taken."
Email	Required, valid email format	"Enter a valid email address"
Email	Must be unique (one account per email)	"Email already registered."
Password	Required, minimum 6 characters	"Password must be at least 6 characters"
Confirm Password	Must exactly match Password	"Passwords do not match."

Note: All validation errors are shown inline on the registration form — the page does not navigate away. Passwords are stored as BCrypt hashes (never plain text).

Default Admin Account

Field	Value
Username	<code>admin</code>
Password	<code>admin123</code>
Role	<code>ROLE_ADMIN</code>

Important: Change the admin password immediately after first login via Admin Portal → Manage Users.

2 Dashboard

The dashboard at `/dashboard` is your home screen after login. It gives an instant overview of all your tasks.

Stat Cards

Card	What it shows
Total	All tasks assigned to you
Pending	Tasks not yet started
Running	Tasks currently in progress
Completed	Tasks marked as done
Failed	Tasks marked as failed
Overdue	Tasks whose end time has passed and are not completed

Panels

- **Upcoming Tasks** — Next 5 tasks ordered by start time
- **Recently Completed** — Last 5 tasks completed

Quick Actions

- **+ New Task** — navigates to the task creation form
- **Manage Categories** — navigates to categories page

Admin view: Admins see stats for ALL users across the system, not just their own tasks.

3 Categories

Categories let you organize tasks by color-coded groups (e.g. "Work", "Personal", "Urgent").

1. Click **Categories** in the left sidebar
2. Enter a category name in the input field
3. Click the color swatch to pick a color (hex color picker)
4. Click **Add Category**
5. Your category appears as a colored card below

Deleting a Category

- Click the ✕ button on any category card
- The category is removed (existing tasks keep their category reference)

Tip: Create categories before creating tasks — you can then assign tasks to them from the task form.

4 Creating a Task

1. Click **My Tasks** in the sidebar
2. Click the **+ Add Task** button (top right)
3. A modal dialog opens — fill in the fields
4. Click **Save Task**
5. A **green toast** confirms the task was saved

Task Form Fields

Field	Required	Description
Task Name	Yes	Short descriptive title
Description	No	Longer details about the task
Priority	Yes	Number 1–10 (10 = highest). Stored in a Max-Heap internally.
Status	Yes	PENDING / RUNNING / COMPLETED / FAILED
Start Time	Yes	When the task begins (date & time picker)
End Time	Yes	When the task ends — must be after start time
Category	No	Pick from your created categories
Recurrence	No	NONE / DAILY / WEEKLY / MONTHLY
Recurrence End Date	Only if recurring	Stop auto-creating after this date
Dependencies	No	Comma-separated IDs of tasks that must complete first

Automatic Checks on Save

- **Overlap Detection** — If another task already occupies the same time slot, save is blocked with an error toast. Uses an Interval Tree ($O(\log n)$).
- **Cycle Detection** — If adding the dependency would create a circular chain ($A \rightarrow B \rightarrow C \rightarrow A$), save is blocked. Uses DFS graph traversal.

5 Managing Tasks

Task List Page (`/tasks`)

Action	How to do it
Search tasks	Type in the search box — filters by name or description instantly
Filter by status	Use the Status dropdown (All / Pending / Running / Completed / Failed)

Filter by category	Use the Category dropdown
Edit a task	Click the  pencil icon on any task row
Delete a task	Click the  trash icon on any task row

Task Status Flow

PENDING → RUNNING → COMPLETED or FAILED

The scheduler automatically transitions tasks to COMPLETED when their end time passes.

6 Recurring Tasks

Recurring tasks automatically create the next occurrence when the current one is completed.

1. In the task form, set **Recurrence** to Daily, Weekly, or Monthly

2. Set a **Recurrence End Date** (the last date a new occurrence will be created)

3. Save the task normally

4. When the scheduler marks it complete, a new task is created automatically with the next date shifted by the recurrence interval

Recurrence Type	Next occurrence created
DAILY	Start + 1 day, End + 1 day
WEEKLY	Start + 7 days, End + 7 days
MONTHLY	Start + 1 month, End + 1 month

 **Note:** The scheduler runs every 60 seconds. Tasks are auto-completed only when their end time is in the past.

7 Task Dependencies

You can specify that a task depends on other tasks being completed first.

1. Find the task IDs you want as dependencies (shown in the task list table)

2. In the task form's **Dependencies** field, enter comma-separated IDs: e.g. `1, 3, 5`

3. The system checks for circular dependencies automatically

Rules

- Dependencies are stored as a directed graph (adjacency list)
- If A depends on B, then B must complete before A

- Circular chains like $A \rightarrow B \rightarrow C \rightarrow A$ are rejected at save time with an error

8 Admin Portal

Only users with **ROLE_ADMIN** can access `/admin`. Regular users get a 403 error.

Admin Dashboard (`/admin`)

- System-wide stats: total users, all tasks, pending, completed, failed, overdue
- Quick links to Manage Users and Manage All Tasks

Manage Users (`/admin/users`)

Action	How
View all users	Full table with username, email, role, status, join date
Change role	Use the Role dropdown (User ↔ Admin) — auto-submits on change
Enable / Disable account	Click the Enable/Disable toggle button — disabled users cannot login

Manage All Tasks (`/admin/tasks`)

Action	How
View every user's tasks	Full task table showing which user owns each task
Search tasks	Type in the search box
Filter by status	Use the Status dropdown
Delete any task	Click the  delete icon on any row

 **Warning:** Deleting a task from the admin panel is permanent and cannot be undone.

9 REST API

The API is available for developers to integrate or automate task management. All endpoints require authentication (same session cookie as the web UI). All responses follow the format:

```
{ "success": true, "data": { ... } }
```

Endpoints

Method	Endpoint	Description
GET	<code>/api/tasks</code>	List your tasks (supports <code>?search=&status=&categoryId=</code>)
POST	<code>/api/tasks</code>	Create a new task

PUT	/api/tasks/{id}	Update a task fully
DELETE	/api/tasks/{id}	Delete a task
PATCH	/api/tasks/{id}/status	Update status only — body: {"status":"COMPLETED"}
GET	/api/tasks/dependencies	Get dependency graph as adjacency map
GET	/api/dashboard/stats	Dashboard statistics as JSON

Example — Create Task

```
POST /api/tasks
Content-Type: application/json

{
  "name": "Write report",
  "priority": 8,
  "status": "PENDING",
  "startTime": "2026-06-10T09:00",
  "endTime": "2026-06-10T11:00"
}
```

10 Registration Validation — Full Reference

All validation runs server-side. Errors are shown inline without leaving the page.

Field	Validation Type	Rule	Error Shown
Username	@NotBlank	Cannot be empty	"Username is required"
Username	@Size	3 to 30 characters	"Username must be 3-30 characters"
Username	DB Unique	Not already registered	"Username already taken."
Email	@NotBlank	Cannot be empty	"Email is required"
Email	@Email	Must be valid format (x@x.x)	"Enter a valid email address"
Email	DB Unique	Not already registered	"Email already registered."
Password	@NotBlank	Cannot be empty	"Password is required"
Password	@Size	Minimum 6 characters	"Password must be at least 6 characters"
Confirm Password	@NotBlank	Cannot be empty	"Please confirm your password"
Confirm Password	Match check	Must equal Password exactly	"Passwords do not match."

 **Security:** Passwords are never stored in plain text. BCrypt hashing with a cost factor of 12 is applied before saving to the database.

Task Scheduler — Work Tracker & Project Manager | Built with Spring Boot 3 | June 2026